

Microlocal Morse Theory and Truncated Microsupport

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Preface

In these notes, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03].

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1 Notations

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a real analytic manifold. We denote the cotangent bundle of X by $\pi: T^*X \rightarrow X$. We denote by $D^b(\mathbf{k}_X)$ the bounded derived category of sheaves of $\mathbf{k}$ -modules on X .

2 Truncated Microsupport

Definition 2.1. Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1 -function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$(2.1) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

2.1 Functorial Properties

Proposition 2.2. Let $f: X \rightarrow Y$ be a morphism of real analytic manifolds and let $F \in D^b(\mathbf{k}_X)$ such that f is proper on $\text{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$(2.2) \quad SS_k(Rf_*F) \subset f_\pi f_d^{-1}(SS_k(F)).$$

The equality holds in case f is a closed embedding.

3 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(3.1) \quad E_n^\bullet: \dots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \dots,$$

and projective limit

$$(3.2) \quad E_\infty^\bullet: \dots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \dots.$$

Denote by

$$(3.3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

Proposition 3.1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $(M_n^j)_n$ satisfies the ML condition. Then

- (a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
- (b) if moreover, for a given i the system $(H^{i-1}(E_n^\bullet))_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 3.2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s, (s \leq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0, s_1} ($s_0 \leq s_1$) are injective (resp. surjective).

ML procedure for sheaves

Proposition 3.3 ([KS90, Proposition 2.7.1]). Let X be a topological space and let $F \in D^+(\mathbf{Z}_X)$. Let $(U_n)_{n \in \mathbf{N}}$ be an increasing sequence of open subsets of X , $(Z_n)_{n \in \mathbf{N}}$ a decreasing sequence of closed subsets of X . Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$.

- (i) For any j , the natural map $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$ is surjective.
- (ii) Assume that for a given j , the projective system $(H_{Z_n}^{j-1}(U_n; F))_n$ satisfies the ML condition. Then φ_j is bijective.
- (iii) Let now $(X_n)_{n \in \mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \text{Int}(X_{n+1})$ for all n . Assume that for a given j , the projective system $(H^{j-1}(X_n; F))_n$ satisfies the ML condition. Then the natural map $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$ is bijective.

4 Results

In this section, let M be a manifold.¹

Theorem 4.1. Let F in $D^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\text{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin \text{SS}_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}]-\infty, t[$. Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each $j < k$.²

¹Can M be of class C^∞ ?

²For $j = k$, this morphism might be injective.

Lemma 4.2. Let $-\infty < a < b < +\infty$ be real numbers and $G \in \mathbf{D}^b(\mathbf{R})$. Let k be an integer. If

$$\left(H_{[t, +\infty[}^j(G) \right)_t \cong 0 \quad \text{for all } t \in [a, b[, \text{ and all } j \leq k.$$

Then one has

$$H^j(-\infty, b[; G) \xrightarrow{\sim} H^j(-\infty, a[; G)$$

for all $j < k$.³

Proof. Let $t \in [a, b[$. We will denote by I_t the open interval $]-\infty, t[$ and denote by Z_t the closed interval $]-\infty, t[(= \overline{I_t})$. By applying the functor $(\cdot)_{\overline{I_t}}$ to the distinguished triangle

$$\mathrm{R}\Gamma_{\mathbf{R}-I_t}(G) \rightarrow G \rightarrow \mathrm{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } \mathbf{D}^b(\mathbf{k}_{\mathbf{R}}),$$

we have

$$(\mathrm{R}\Gamma_{[t, +\infty[}(G))_{\{t\}} \rightarrow G_{Z_t} \rightarrow \mathrm{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } \mathbf{D}^b(\mathbf{k}_{\mathbf{R}}).$$

Applying the functor $\mathrm{R}\Gamma(\mathbf{R}; \cdot)$ to this, we have

$$(4.1) \quad (\mathrm{R}\Gamma_{[t, +\infty[}(G))_t \rightarrow \mathrm{R}\Gamma(Z_t; G) \rightarrow \mathrm{R}\Gamma(I_t; G) \xrightarrow{+1} \quad \text{in } \mathbf{D}^b(\mathbf{k}).$$

Then consider the following conditions for $j \in \mathbf{Z}$, $s \in [a, b[$, and $t \in]a, b[$:

$$\begin{aligned} (a)_s^j & \quad \varinjlim_{t>s} H^j(I_t; G) \xrightarrow{\sim} H^j(I_s; G), \\ (b)_t^j & \quad \varprojlim_{s<t} H^j(I_t; G) \xleftarrow{\sim} H^j(I_t; G). \end{aligned}$$

By hypothesis, it follows from (11.1) that $H^j(Z_t; G) \xrightarrow{\sim} H^j(I_t; G)$ for $t \in [a, b[$.⁴ Since $\varinjlim_{t>s} H^j(I_t; G) \cong H^j(Z_s; G)$, $(a)_s^j$ holds for any $j < k$ and $s \in [a, b[$.⁵

Moreover, since we assume G is a bounded complex, $(b)_t^j$ holds for $j \ll 0$ and for all $t \in]a, b[$. Let us denote such j by j_0 . Let us argue by induction on j . Assume $(b)_t^j$ holds for all $t \in]a, b[$ and all $j \leq j_0$. Applying Proposition 10.2, we find the isomorphisms

$$(4.2) \quad H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j \leq j_0 \text{ and all } a < s \leq t < b.$$

Let $G \xrightarrow{\sim} G^\bullet$ be a flabby resolution. Let $t \in]a, b[$. Consider a complex

$$E_n^\bullet = \Gamma(I_{t-\frac{1}{n}}; G^\bullet).$$

³For k , this morphism might be injective. (not written in this draft.)

⁴Similarly, we have injections $H^k(Z_t; G) \hookrightarrow H^k(I_t; G)$ for $t \in [a, b[$.

⁵Similarly,

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(Z_s; G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G)$$

for all $s \in [a, b[$.

Since $G^\bullet$ is flabby, we have $E_n^j \cong E_\infty^j$ and the projective system $(E_n^j)_n$ satisfies the ML condition for all $j \in \mathbf{Z}$.

By (11.2), the projective system $(H^{j_0}(E_n^\bullet))_n$ satisfies the ML condition (because we have $a < t - \frac{1}{n} < t$ for $n \gg 0$). Applying Proposition 10.1 to $(E_n^\bullet)_n$, it follows that $(b)_t^j$ holds for $j_0 + 1$ and the induction proceeds. More precisely, we have

$$\begin{aligned} H^{j_0+1}(I_t; G^\bullet) &\xrightarrow{\sim} \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G^\bullet) \\ &\cong \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G) \\ &\cong \varprojlim_{s < t} H^{j_0+1}(I_s; G). \end{aligned}$$

Again by Proposition 10.2, we get the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j < k \text{ and all } a < s \leq t \leq b,$$

and injections

$$H^j(I_s; G) \hookrightarrow H^j(I_t; G)$$

for $j = k$ and for all $a < s \leq t \leq b$.

Finally, using (11.1) and the hypothesis, we have the isomorphisms

$$H^j(I_a; G) \xleftarrow{\sim} H^j(Z_a; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } j < k \text{ and all } a < s \leq b.$$

Similarly, for $j = k$, we have

$$H^k(I_b; G) \cong \varinjlim_{t > s} H^k(I_t; G) \hookrightarrow H^k(I_s; G) \hookrightarrow H^k(I_a; G).$$

□

Proof of Theorem 11.1. First, we have

$$H^j(M_t; F) \cong H^j \mathrm{R}\Gamma(\text{]}-\infty, t[; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leq \psi(x) < b$, and by (9.2), the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \mathrm{SS}_k(F)$ holds. Hence we have

$$(H^j \mathrm{R}\Gamma_{[\psi(x), +\infty[} \mathrm{R}\psi_* F)_{\psi(x)} = 0$$

for each $j \leq k$. Then we can apply Lemma 11.2 and have

$$H^j(\text{]}-\infty, b[; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j(\text{]}-\infty, a[; \mathrm{R}\psi_* F)$$

for all $j < k$. Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

for all $j < k$.

□

5 Truncated Noncharacteristic Deformation

We will reformulate the noncharacteristic deformation lemma [KS90, Proposition 2.7.2] to make a relation to truncated microsupport.

Proposition 5.1. Let X be a Hausdorff space, $F \in \mathcal{D}^b(\mathbf{Z}_X)$, and let $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t - U_s} \cap \text{supp } F$ is compact.
- (iii) Setting $Z_s := \bigcap_{t > s} \overline{U_t - U_s}$, we have for all pairs (s, t) with $s \leq t$, and all $x \in Z_s - U_t$,
$$(H_{X - U_t}^j(F))_x = 0 \quad \text{for all } j \leq k.$$
- (iv) For any $s \in \mathbf{R}$, $\bigcap_{t > s} U_t = \overline{U_s}$ and $\overline{U_s}$ is closed in a paracompact open subset of X .
- (v) ⁶ $Z_s - U_s \subset \partial U_s$ for all $s \in \mathbf{R}$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j \left(\bigcup_s U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left(\bigcup_s U_s; F \right) \hookrightarrow H^k(U_t; F).$$

Proof. Consider the following conditions for $j \in \mathbf{Z}$, s , and t :

$$\begin{aligned} (a)_s^j & \quad \mu_s^j: \varinjlim_{t > s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (a')_s^k & \quad \mu_s^k: \varinjlim_{t > s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\ (b)_s^j & \quad \lambda_s^j: H^j(U_t; F) \xrightarrow{\sim} \varprojlim_{s < t} H^j(U_s; F). \end{aligned}$$

Let us prove $(a)_s^j$ for $s \in \mathbf{R}$ and $j < k$. We can assume that $\overline{U_t - U_s}$ is compact for any $s \leq t$ by replacing X by $\text{supp } F$. Consider the distinguished triangle

$$\text{R}\Gamma_{X - U_t}(F) \rightarrow F \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Applying to this triangle the functor $(\cdot)_{\overline{U_t}}$, we have

$$(\text{R}\Gamma_{X - U_t}(F))_{\overline{U_t}} \rightarrow F_{\overline{U_t}} \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Again applying to this triangle the functor $\text{R}\Gamma(X; \cdot)$, we have

$$(5.1) \quad \text{R}\Gamma(X; (\text{R}\Gamma_{X - U_t}(F))_{\overline{U_t}}) \rightarrow \text{R}\Gamma(\overline{U_t}; F) \rightarrow \text{R}\Gamma(U_t; F) \xrightarrow{+1}.$$

⁶This condition might be deduced from (iii).

By hypotheses (iii)–(v), it follows that

$$\left(H_{X \dashrightarrow U_t}^j(F)\right)_{\partial U_t} = 0$$

for any $j \leq k$ and then

$$H^j(X; (\mathrm{R}\Gamma_{X \dashrightarrow U_t}(F))_{\overline{U_t}}) = 0 \quad \text{for any } j \leq k.$$

⁷Thus for any t , we have

$$(5.2) \quad H^j(\overline{U_t}; F) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for any } j < k,$$

$$(5.3) \quad H^k(\overline{U_t}; F) \hookrightarrow H^k(U_t; F).$$

Therefore, we have by (iv) and (12.2),

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j\left(\bigcap_{t>s} U_t; F\right) \\ &= H^j(\overline{U_s}; F) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have by (iv) and (12.3),

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k\left(\bigcap_{t>s} U_t; F\right) \\ &= H^k(\overline{U_s}; F) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and $(a')_s^k$ follows. □

⁷この辺の議論はきちんと詰めないといけない。

Remark 5.2. Lemma 11.2 follows from Proposition 12.1. Indeed, putting $X =]-\infty, b[$, $F = G|_{I_b}$, and $U_t = I_t$, we find that

- (i) $\bigcup_{s < t} I_s = \bigcup_{s < t}]-\infty, s[=]-\infty, t[= I_t$ for all $t \in \mathbf{R}$.
- (ii) If $s \leq t$, then $\overline{I_t - I_s} \cap \text{supp } G = \overline{[s, t[} \cap \text{supp } G = [s, t] \cap \text{supp } G$ is compact.
- (iii) For $s \in \mathbf{R}$, we have

$$Z_s = \bigcap_{t > s} \overline{I_t - I_s} = \bigcap_{t > s} \overline{[s, t[} = \bigcap_{t > s} [s, t] = \{s\},$$

and

$$Z_s - I_t = \{s\} -]-\infty, t[= \begin{cases} \{s\} & \text{if } s = t, \\ \emptyset & \text{if } s < t. \end{cases}$$

By hypothesis of Lemma 11.2, for all $s \in \{s\}$,

$$H_{X - I_s}^j(F)_s = H_{[s, \infty[}^j(F)_s = 0.$$

Then we can apply Proposition 12.1, and we have the isomorphisms for all $t \in [a, b[$ and $j < k$,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the injections

$$H^k(I_b; F) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F)$$

6 Relative Microlocal Morse Theory via Truncated Microsupport

Let $f: Y \rightarrow X$ be a morphism of manifolds, and φ a real valued function of class C^1 . For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &:= \{y \in Y ; \varphi(y) < t\} = \varphi^{-1}(]-\infty, t[), \\ \tilde{Y}_t &:= \{y \in Y ; \varphi(y) \leq t\} = \varphi^{-1}(]-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

Proposition 6.1. Let $G \in \mathcal{D}^b(Y)$, and assume $f_t: \text{supp}(G) \cap \overline{Y_t} \rightarrow X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and t_o a real number.

(i) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned} (a) \quad & R^j f_* G \cong R^j f_{t*} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\ (a_k) \quad & R^k f_* G \hookrightarrow R^k f_{t*} j_t^{-1} G \quad \text{for } t > t_o, \\ (a') \quad & R^j \tilde{f}_* G \cong R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } j < k \text{ and } t \geq t_o, \\ (a'_k) \quad & R^k \tilde{f}_* G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } t \geq t_o, \\ (b) \quad & \text{SS}_k(Rf_* G) \subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ & = f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

(ii) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$-d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned} (c) \quad & R^j f_! G \cong R^j f_{t!} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\ (c_k) \quad & R^k f_! G \hookrightarrow R^k f_{t!} j_t^{-1} G \quad \text{for } t > t_o, \\ (c') \quad & R^j \tilde{f}_! G \cong R^j \tilde{f}_{t*} \tilde{j}_t^! G \quad \text{for } j < k \text{ and } t \geq t_o, \\ (c'_k) \quad & R^k \tilde{f}_! G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^! G \quad \text{for } t \geq t_o, \\ (d) \quad & \text{SS}_k(Rf_! G) \subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ & = f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

Proof. Let $\tilde{f}$ be the morphism $(f, \varphi): Y \rightarrow X \times \mathbf{R}$, and let q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. Then $\tilde{f}$ is proper on $\text{supp}(G)$ and $f = q \circ \tilde{f}$.⁸ Let i_t be the open embedding $X \times]-\infty, t[\hookrightarrow X \times \mathbf{R}$ and $\tilde{i}_t$ the closed embedding $X \times]-\infty, t] \hookrightarrow X \times \mathbf{R}$. Set $q_t = q \circ i_t$ and $\tilde{q}_t = q \circ \tilde{i}_t$. These morphisms can be visualized by the following diagrams.

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X, \\ \uparrow j_t & & \uparrow i_t \searrow q_t \\ Y_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t[\end{array} \quad \begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X. \\ \uparrow \tilde{j}_t & & \uparrow \tilde{i}_t \searrow \tilde{q}_t \\ Y_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t] \end{array}$$

Then, setting $\tilde{G} := Rf_* G$, we have

$$\begin{aligned} Rf_* G &\cong Rq_* \tilde{G}, & R\tilde{f}_* G &\cong R\tilde{q}_* \tilde{G}, \\ Rf_{t*} j_t^{-1} G &\cong Rq_{t*} i_t^{-1} \tilde{G}, & R\tilde{f}_{t*} \tilde{j}_t^{-1} G &\cong R\tilde{q}_{t*} \tilde{i}_t^{-1} \tilde{G}, \\ Rf_{t!} j_t^{-1} G &\cong Rq_{t!} i_t^{-1} \tilde{G}, & R\tilde{f}_{t*} \tilde{j}_t^! G &\cong R\tilde{q}_{t*} \tilde{i}_t^! \tilde{G}. \end{aligned}$$

⁸This can be proved as follows. If K is a compact set of $X \times \mathbf{R}$, $\tilde{f}^{-1}(K)$ is compact.

Thus taking j -th cohomology, we have

$$\begin{aligned} R^j f_* G &\cong R^j q_* \tilde{G}, & R^j \tilde{f}_* G &\cong R^j \tilde{q}_* \tilde{G}, \\ R^j f_{t*} j_t^{-1} G &\cong R^j q_{t*} i_t^{-1} \tilde{G}, & R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G &\cong R^j \tilde{q}_{t*} \tilde{i}_t^{-1} \tilde{G}, \\ R^j f_{t!} j_t^{-1} G &\cong R^j q_{t!} i_t^{-1} \tilde{G}, & R^j \tilde{f}_{t*} \tilde{j}_t^! G &\cong R^j \tilde{q}_{t*} \tilde{i}_t^! \tilde{G}. \end{aligned}$$

Therefore, it is enough to prove the result for $\tilde{G}$, q , i_t , $\tilde{\varphi}$ instead of G , f , j_t , φ .
(i) □

7 Preface

In this thesis, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03]. In [KFS03], they do not study the functorial property of truncated microsupport of direct images of non proper morphisms. We prove this functorial property under non proper direct images.

8 Notations and Reviews

We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a topological space. We denote by $D^b(\mathbf{k}_X)$ the bounded derived category of sheaves of $\mathbf{k}$ -modules on X .

Let us denote Z a locally closed set of X and $i: Z \hookrightarrow X$ an inclusion. We use the following functors:

$$R\Gamma_Z(\cdot) := Ri_* i^!, \quad (\cdot)_Z := Ri_! i^{-1}.$$

Geometrical Notations

Let M be a real manifold of class C^∞ . We denote by $\pi: T^*M \rightarrow M$ the cotangent bundle of M . If $f: M \rightarrow N$ is a morphism of manifolds, then we have the natural diagram.

$$\begin{array}{ccc} TM & \xrightarrow{f'} M \times_N TN & \xrightarrow{f_\tau} TN \\ & \searrow \tau_M \downarrow & \downarrow \tau_N \\ & M & \xrightarrow{f} N, \end{array} \quad \begin{array}{ccc} T^*M & \xleftarrow{f_d} M \times_N T^*N & \xrightarrow{f_\pi} T^*N \\ & \searrow \pi_M \downarrow & \downarrow \pi_N \\ & M & \xrightarrow{f} N. \end{array}$$

In this diagram, f_τ and f_π denote the natural projections. f' is defined by $(x; v) \mapsto (x; df_x v)$, and f_d is induced by the transposition of f' .

9 Truncated Microsupport

Definition 9.1. Let $F \in D^b(\mathbf{k}_X)$. The closed conic subset $SS_k(F)$ of T^*X is defined by: $p \notin SS_k(F)$ if and only if there exists an open conic neighborhood U

of p such that for any $x \in \pi(U)$ and for any real-valued C^1 -function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$(9.1) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

Functorial Properties

Proposition 9.2. Let $f: X \rightarrow Y$ be a morphism of real analytic manifolds and let $F \in D^b(\mathbf{k}_X)$ such that f is proper on $\text{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$(9.2) \quad \text{SS}_k(Rf_*F) \subset f_{\pi} f_d^{-1}(\text{SS}_k(F)).$$

The equality holds in case f is a closed embedding.

10 Lemmas from [KS90]

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_{\infty} := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(10.1) \quad E_n^{\bullet}: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and projective limit

$$(10.2) \quad E_{\infty}^{\bullet}: \cdots \rightarrow M_{\infty}^{j-1} \rightarrow M_{\infty}^j \rightarrow M_{\infty}^{j+1} \rightarrow \cdots.$$

Denote by

$$(10.3) \quad \Phi_k: H^k(E_{\infty}^{\bullet}) \rightarrow \varprojlim_n H^k(E_n^{\bullet})$$

the natural morphism.

Proposition 10.1 ([KS90, Proposition 1.12.4]). Assume that for each $j \in \mathbf{Z}$, the system $(M_n^j)_n$ satisfies the ML condition. Then

- (a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
- (b) if moreover, for a given i the system $(H^{i-1}(E_n^{\bullet}))_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 10.2 ([KS90, Proposition 1.12.6]). Let $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s, (s \leq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0, s_1} ($s_0 \leq s_1$) are injective (resp. surjective).

ML procedure for sheaves

Proposition 10.3 ([KS90, Proposition 2.7.1]). Let X be a topological space and let $F \in \mathbf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n \in \mathbf{N}}$ be an increasing sequence of open subsets of X , $(Z_n)_{n \in \mathbf{N}}$ a decreasing sequence of closed subsets of X . Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$.

- (i) For any j , the natural map $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$ is surjective.
- (ii) Assume that for a given j , the projective system $(H_{Z_n}^{j-1}(U_n; F))_n$ satisfies the ML condition. Then φ_j is bijective.
- (iii) Let now $(X_n)_{n \in \mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \text{Int}(X_{n+1})$ for all n . Assume that for a given j , the projective system $(H^{j-1}(X_n; F))_n$ satisfies the ML condition. Then the natural map $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$ is bijective.

11 Results

In this section, let M be a manifold.⁹

Theorem 11.1. Let F in $\mathbf{D}^b(\mathbf{k}_M)$, and $\psi: M \rightarrow \mathbf{R}$ be a real-valued function of class C^1 , proper on $\text{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin \text{SS}_k(F)$ for $a \leq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}(]-\infty, t])$. Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each $j < k$.¹⁰

Lemma 11.2. Let $-\infty < a < b < +\infty$ be real numbers and $G \in \mathbf{D}^b(\mathbf{R})$. Let k be an integer. If

$$\left(H_{[t, +\infty[}^j(G) \right)_t \cong 0 \quad \text{for all } t \in [a, b[, \text{ and all } j \leq k.$$

Then one has

$$H^j(]-\infty, b[; G) \xrightarrow{\sim} H^j(]-\infty, a[; G)$$

for all $j < k$.¹¹

Proof. Let $t \in [a, b[$. We will denote by I_t the open interval $]-\infty, t[$ and denote by Z_t the closed interval $]-\infty, t](= \overline{I_t})$. By applying the functor $(\cdot)_{\overline{I_t}}$ to the distinguished triangle

$$\text{R}\Gamma_{\mathbf{R}-I_t}(G) \rightarrow G \rightarrow \text{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } \mathbf{D}^b(\mathbf{k}_{\mathbf{R}}),$$

we have

$$(\text{R}\Gamma_{[t, +\infty[}(G))_{\{t\}} \rightarrow G_{Z_t} \rightarrow \text{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } \mathbf{D}^b(\mathbf{k}_{\mathbf{R}}).$$

⁹Can M be of class C^∞ ?

¹⁰For $j = k$, this morphism might be injective.

¹¹For k , this morphism might be injective. (not written in this draft.)

Applying the functor $\mathrm{R}\Gamma(\mathbf{R}; \cdot)$ to this, we have

$$(11.1) \quad (\mathrm{R}\Gamma_{[t, +\infty[}(G))_t \rightarrow \mathrm{R}\Gamma(Z_t; G) \rightarrow \mathrm{R}\Gamma(I_t; G) \xrightarrow{+1} \quad \text{in } \mathbf{D}^b(\mathbf{k}).$$

Then consider the following conditions for $j \in \mathbf{Z}$, $s \in [a, b[$, and $t \in]a, b]$:

$$\begin{aligned} (a)_s^j & \quad \varinjlim_{t>s} H^j(I_t; G) \xrightarrow{\sim} H^j(I_s; G), \\ (b)_t^j & \quad \varprojlim_{s<t} H^j(I_t; G) \xleftarrow{\sim} H^j(I_t; G). \end{aligned}$$

By hypothesis, it follows from (11.1) that $H^j(Z_t; G) \xrightarrow{\sim} H^j(I_t; G)$ for $t \in [a, b]$.¹² Since $\varinjlim_{t>s} H^j(I_t; G) \cong H^j(Z_s; G)$, $(a)_s^j$ holds for any $j < k$ and $s \in [a, b]$.¹³

Moreover, since we assume G is a bounded complex, $(b)_t^j$ holds for $j \ll 0$ and for all $t \in]a, b]$. Let us denote such j by j_0 . Let us argue by induction on j . Assume $(b)_t^j$ holds for all $t \in]a, b[$ and all $j \leq j_0$. Applying Proposition 10.2, we find the isomorphisms

$$(11.2) \quad H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j \leq j_0 \text{ and all } a < s \leq t < b.$$

Let $G \xrightarrow{\sim} G^\bullet$ be a flabby resolution. Let $t \in]a, b]$. Consider a complex

$$E_n^\bullet = \Gamma(I_{t-\frac{1}{n}}; G^\bullet).$$

Since $G^\bullet$ is flabby, we have $E_n^j \cong E_\infty^j$ and the projective system $(E_n^j)_n$ satisfies the ML condition for all $j \in \mathbf{Z}$.

By (11.2), the projective system $(H^{j_0}(E_n^\bullet))_n$ satisfies the ML condition (because we have $a < t - \frac{1}{n} < t$ for $n \gg 0$). Applying Proposition 10.1 to $(E_n^\bullet)_n$, it follows that $(b)_t^{j_0}$ holds for $j_0 + 1$ and the induction proceeds. More precisely, we have

$$\begin{aligned} H^{j_0+1}(I_t; G^\bullet) & \xrightarrow{\sim} \varprojlim_{\Phi_{j_0+1}} \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G^\bullet) \\ & \cong \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G) \\ & \cong \varprojlim_{s<t} H^{j_0+1}(I_s; G). \end{aligned}$$

Again by Proposition 10.2, we get the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j < k \text{ and all } a < s \leq t \leq b,$$

and injections

$$H^j(I_s; G) \hookrightarrow H^j(I_t; G)$$

¹²Similarly, we have injections $H^k(Z_t; G) \hookrightarrow H^k(I_t; G)$ for $t \in [a, b]$.

¹³Similarly,

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(Z_s; G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G)$$

for all $s \in [a, b]$.

for $j = k$ and for all $a < s \leq t \leq b$.

Finally, using (11.1) and the hypothesis, we have the isomorphisms

$$H^j(I_a; G) \xleftarrow{\sim} H^j(Z_a; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } j < k \text{ and all } a < s \leq b.$$

Similarly, for $j = k$, we have

$$H^k(I_b; G) \cong \varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G) \hookrightarrow H^k(I_a; G).$$

□

Proof of Theorem 11.1. First, we have

$$H^j(M_t; F) \cong H^j \operatorname{R}\Gamma(\cdot, t[; \operatorname{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \operatorname{SS}_k(F)$ for $a \leq \psi(x) < b$, and by (9.2), the formula $\operatorname{SS}_k(\operatorname{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \operatorname{SS}_k(F)$ holds. Hence we have

$$(H^j \operatorname{R}\Gamma_{[\psi(x), +\infty[} \operatorname{R}\psi_* F)_{\psi(x)} = 0$$

for each $j \leq k$. Then we can apply Lemma 11.2 and have

$$H^j(\cdot, -\infty, b[; \operatorname{R}\psi_* F) \xrightarrow{\sim} H^j(\cdot, -\infty, a[; \operatorname{R}\psi_* F)$$

for all $j < k$. Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

for all $j < k$. □

12 Truncated Noncharacteristic Deformation

We will reformulate the noncharacteristic deformation lemma [KS90, Proposition 2.7.2] to make a relation to truncated microsupport.

Proposition 12.1. Let X be a Hausdorff space, $F \in \mathcal{D}^b(\mathbf{Z}_X)$, and let $(U_t)_{t \in \mathbf{R}}$ be a family of open subsets of X . We assume the following conditions.

- (i) $U_t = \bigcup_{s < t} U_s$, for all $t \in \mathbf{R}$.
- (ii) For all pairs (s, t) with $s \leq t$, the set $\overline{U_t - U_s} \cap \operatorname{supp} F$ is compact.
- (iii) Setting $Z_s := \bigcap_{t > s} \overline{U_t - U_s}$, we have for all pairs (s, t) with $s \leq t$, and all $x \in Z_s - U_t$,
$$(H_{X - U_t}^j(F))_x = 0 \quad \text{for all } j \leq k.$$
- (iv) For any $s \in \mathbf{R}$, $\bigcap_{t > s} U_t = \overline{U_s}$ and $\overline{U_s}$ is closed in a paracompact open subset of X .
- (v) ¹⁴ $Z_s - U_s \subset \partial U_s$ for all $s \in \mathbf{R}$.

¹⁴This condition might be deduced from (iii).

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j \left(\bigcup_s U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left(\bigcup_s U_s; F \right) \hookrightarrow H^k(U_t; F).$$

Proof. Consider the following conditions for $j \in \mathbf{Z}$, s , and t :

$$\begin{aligned} (a)_s^j & \quad \mu_s^j: \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (a')_s^k & \quad \mu_s^k: \varinjlim_{t>s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\ (b)_s^j & \quad \lambda_s^j: H^j(U_t; F) \xrightarrow{\sim} \varprojlim_{s<t} H^j(U_s; F). \end{aligned}$$

Let us prove $(a)_s^j$ for $s \in \mathbf{R}$ and $j < k$. We can assume that $\overline{U_t - U_s}$ is compact for any $s \leq t$ by replacing X by $\text{supp } F$. Consider the distinguished triangle

$$\text{R}\Gamma_{X \dashv U_t}(F) \rightarrow F \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Applying to this triangle the functor $(\cdot)_{\overline{U_t}}$, we have

$$(\text{R}\Gamma_{X \dashv U_t}(F))_{\overline{U_t}} \rightarrow F_{\overline{U_t}} \rightarrow \text{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Again applying to this triangle the functor $\text{R}\Gamma(X; \cdot)$, we have

$$(12.1) \quad \text{R}\Gamma(X; (\text{R}\Gamma_{X \dashv U_t}(F))_{\overline{U_t}}) \rightarrow \text{R}\Gamma(\overline{U_t}; F) \rightarrow \text{R}\Gamma(U_t; F) \xrightarrow{+1}.$$

By hypotheses (iii)-(v), it follows that

$$\left(H_{X \dashv U_t}^j(F) \right)_{\partial U_t} = 0$$

for any $j \leq k$ and then

$$H^j(X; (\text{R}\Gamma_{X \dashv U_t}(F))_{\overline{U_t}}) = 0 \quad \text{for any } j \leq k.$$

¹⁵Thus for any t , we have

$$(12.2) \quad H^j(\overline{U_t}; F) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for any } j < k,$$

$$(12.3) \quad H^k(\overline{U_t}; F) \hookrightarrow H^k(U_t; F).$$

Therefore, we have by (iv) and (12.2),

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) & \cong H^j \left(\bigcap_{t>s} U_t; F \right) \\ & = H^j(\overline{U_s}; F) \\ & \xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

¹⁵ この辺の議論はきちんと詰めないといけない。

for any $j < k$, and $(a)_s^j$ follows. Similarly, we have by (iv) and (12.3),

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k\left(\bigcap_{t>s} U_t; F\right) \\ &= H^k(\overline{U_s}; F) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and $(a')_s^k$ follows. $\square$

Remark 12.2. Lemma 11.2 follows from Proposition 12.1. Indeed, putting $X =]-\infty, b[$, $F = G|_{I_b}$, and $U_t = I_t$, we find that

- (i) $\bigcup_{s<t} I_s = \bigcup_{s<t}]-\infty, s[=]-\infty, t[= I_t$ for all $t \in \mathbf{R}$.
- (ii) If $s \leq t$, then $\overline{I_t - I_s} \cap \text{supp } G = \overline{[s, t[} \cap \text{supp } G = [s, t] \cap \text{supp } G$ is compact.
- (iii) For $s \in \mathbf{R}$, we have

$$Z_s = \bigcap_{t>s} \overline{I_t - I_s} = \bigcap_{t>s} \overline{[s, t[} = \bigcap_{t>s} [s, t] = \{s\},$$

and

$$Z_s - I_t = \{s\} -]-\infty, t[= \begin{cases} \{s\} & \text{if } s = t, \\ \emptyset & \text{if } s < t. \end{cases}$$

By hypothesis of Lemma 11.2, for all $s \in \{s\}$,

$$H_{X - I_s}^j(F)_s = H_{[s, \infty[}^j(F)_s = 0.$$

Then we can apply Proposition 12.1, and we have the isomorphisms for all $t \in [a, b[$ and $j < k$,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the injections

$$H^k(I_b; F) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F)$$

13 Relative Microlocal Morse Theory via Truncated Microsupport

Let $f: Y \rightarrow X$ be a morphism of manifolds, and φ a real valued function of class C^1 . For a real number $t \in \mathbf{R}$, we set

$$\begin{aligned} Y_t &:= \{y \in Y ; \varphi(y) < t\} = \varphi^{-1}(]-\infty, t[), \\ \tilde{Y}_t &:= \{y \in Y ; \varphi(y) \leq t\} = \varphi^{-1}(]-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

Proposition 13.1. Let $G \in \mathbf{D}^b(Y)$, and assume $f_t: \text{supp}(G) \cap \overline{Y_t} \rightarrow X$ is proper for all $t \in \mathbf{R}$. Let k be an integer and t_o a real number.

(i) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

- (a) $R^j f_* G \cong R^j f_{t*} j_t^{-1} G$ for $j < k$ and $t > t_o$,
- (a_k) $R^k f_* G \hookrightarrow R^k f_{t*} j_t^{-1} G$ for $t > t_o$,
- (a') $R^j \tilde{f}_* G \cong R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G$ for $j < k$ and $t \geq t_o$,
- (a'_k) $R^k \tilde{f}_* G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^{-1} G$ for $t \geq t_o$,
- (b) $\text{SS}_k(Rf_* G) \subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right)$
 $= f_\pi f_d^{-1} (\text{SS}_k(G)).$

(ii) Assume that, for any $y \in Y - \tilde{Y}_{t_o}$, we have

$$-d\varphi(y) \notin \left(\text{SS}_k(G) + f_d \left(Y \times_X T^*X \right) \right).$$

Then

- (c) $R^j f_! G \cong R^j f_{t!} j_t^{-1} G$ for $j < k$ and $t > t_o$,
- (c_k) $R^k f_! G \hookrightarrow R^k f_{t!} j_t^{-1} G$ for $t > t_o$,
- (c') $R^j \tilde{f}_! G \cong R^j \tilde{f}_{t!} \tilde{j}_t^{-1} G$ for $j < k$ and $t \geq t_o$,
- (c'_k) $R^k \tilde{f}_! G \hookrightarrow R^k \tilde{f}_{t!} \tilde{j}_t^{-1} G$ for $t \geq t_o$,
- (d) $\text{SS}_k(Rf_! G) \subset f_\pi \left(f_d^{-1} \left(\text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right)$
 $\subset f_\pi f_d^{-1} (\text{SS}_k(G)).$

Proof. Let $\tilde{f}$ be the morphism $(f, \varphi): Y \rightarrow X \times \mathbf{R}$, and let q and $\tilde{\varphi}$ be the first and second projections defined on $X \times \mathbf{R}$ respectively. Then $\tilde{f}$ is proper on $\text{supp}(G)$ and $f = q \circ \tilde{f}$.¹⁶ Let i_t be the open embedding $X \times]-\infty, t[\hookrightarrow X \times \mathbf{R}$ and $\tilde{i}_t$ the closed embedding $X \times]-\infty, t] \hookrightarrow X \times \mathbf{R}$. Set $q_t = q \circ i_t$ and $\tilde{q}_t = q \circ \tilde{i}_t$. These morphisms can be visualized by the following diagrams.

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X, \\ j_t \uparrow & & \uparrow i_t \nearrow q_t \\ \tilde{Y}_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t[\end{array} \qquad \begin{array}{ccc} Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} \xrightarrow{q} X, \\ \tilde{j}_t \uparrow & & \uparrow \tilde{i}_t \nearrow \tilde{q}_t \\ \tilde{Y}_t & \xrightarrow{\tilde{f}_t} & X \times]-\infty, t] \end{array}$$

¹⁶This can be proved as follows. If K is a compact set of $X \times \mathbf{R}$, $\tilde{f}^{-1}(K)$ is compact.

Then, setting $\tilde{G} := Rf_*G$, we have

$$\begin{aligned} Rf_*G &\cong Rq_*\tilde{G}, & R\tilde{f}_*G &\cong R\tilde{q}_*\tilde{G}, \\ Rf_{t*}j_t^{-1}G &\cong Rq_{t*}i_t^{-1}\tilde{G}, & R\tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong R\tilde{q}_{t*}\tilde{i}_t^{-1}\tilde{G}, \\ Rf_{t!}j_t^{-1}G &\cong Rq_{t!}i_t^{-1}\tilde{G}, & R\tilde{f}_{t*}\tilde{j}_t^!G &\cong R\tilde{q}_{t*}\tilde{i}_t^!\tilde{G}. \end{aligned}$$

Thus taking j -th cohomology, we have

$$\begin{aligned} R^j f_*G &\cong R^j q_*\tilde{G}, & R^j \tilde{f}_*G &\cong R^j \tilde{q}_*\tilde{G}, \\ R^j f_{t*}j_t^{-1}G &\cong R^j q_{t*}i_t^{-1}\tilde{G}, & R^j \tilde{f}_{t*}\tilde{j}_t^{-1}G &\cong R^j \tilde{q}_{t*}\tilde{i}_t^{-1}\tilde{G}, \\ R^j f_{t!}j_t^{-1}G &\cong R^j q_{t!}i_t^{-1}\tilde{G}, & R^j \tilde{f}_{t*}\tilde{j}_t^!G &\cong R^j \tilde{q}_{t*}\tilde{i}_t^!\tilde{G}. \end{aligned}$$

Therefore, it is enough to prove the result for $\tilde{G}$, q , i_t , $\tilde{\varphi}$ instead of G , f , j_t , φ .

(i) It follows from the hypothesis that

$$(13.1) \quad \text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\}) \subset T^*X \times \{(t; \tau); \tau \leq 0\}.$$

Indeed, by hypothesis, for any $(y; \eta) \in \text{SS}_k(G)$ and $(f(x); \xi) \in Y \times_X T^*X$, we have

$$(13.2) \quad d\varphi(y) \neq \eta + f^*\xi.$$

If $(x, t; \xi, \tau)$ is a point of $\text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\})$, then, by the formula (9.2), there exists $(y; \eta) \in \text{SS}_k(G)$ such that

$$\tilde{f}(y) = (x, t) \text{ and } \eta = \tilde{f}^*(\xi, \tau).$$

For this η , we have

$$\eta = \tilde{f}^*(\xi, \tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If $\tau > 0$, we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau}f^*\xi$$

and this contradicts (13.2) because $\text{SS}_k(G)$ is conic.

By (13.1), Let U be an open subset of X contained in a local chart, and convex in such a chart. By Proposition 12.1, we have isomorphisms for all $t > t_o$ and $j < k$

$$(13.3) \quad H^j(U \times \mathbf{R}; \tilde{G}) \xrightarrow{\sim} H^j(U \times]-\infty, t[; \tilde{G})$$

and have an injection for all $t > t_o$

$$(13.4) \quad H^k(U \times \mathbf{R}; \tilde{G}) \hookrightarrow H^k(U \times]-\infty, t[; \tilde{G}).$$

This can be showed as follows.

This proves the result. □

Remark 13.2. $(U \times I_t)$ is not a gentle family.

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